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Systems and methods for evaluation of vehicle parameters of a remotely controllable material handling vehicle

Abstract

A system for a remotely controllable material handling vehicle switchable between a manual mode and a travel request mode is provided. The system can include a controller including a processor and a memory, where vehicle condition data previously evaluated by the controller can be stored in the memory. A remote control device can be in communication with the controller, where a travel control function on the remote control device can be configured to provide a travel request to the controller to cause the material handling vehicle to move forward when the material handling vehicle is in the travel request mode. In some aspects, the controller can be configured to receive the travel request, recall at least one previously evaluated vehicle condition data stored in the memory, and, based on the at least one vehicle condition data stored in the memory, command the material handling vehicle to move forward.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is based on and claims the priority to U.S. Provisional Patent Application No. 62/986,299, filed on Mar. 6, 2020, and entitled “Systems and Methods for Evaluation of Vehicle Parameters of a Remotely Controllable Material Handling Vehicle.”

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH
(1) Not Applicable.
BACKGROUND
(2) Warehouses typically employ the use of material handling vehicles, specifically, operators may use a remote control device to control travel of a material handling vehicle within the warehouse.
BRIEF SUMMARY
(3) The present disclosure relates generally to material handling vehicles and, more specifically, to remotely controllable material handling vehicles that can be switched between a manual operation mode and a travel request mode.
(4) According to some aspects of the present disclosure, a system for a remotely controllable material handling vehicle switchable between a manual mode and a travel request mode is provided. The system can include a controller including a processor and a memory, where vehicle condition data previously evaluated by the controller can be stored in the memory. In some aspects, a remote control device can be in communication with the controller, where a travel control function on the remote control device can be configured to provide a travel request to the controller to cause the material handling vehicle to move forward when the material handling vehicle is in the travel request mode. In some aspects, the controller can be configured to receive the travel request, recall at least one previously evaluated vehicle condition data stored in the memory, and, based on the at least one vehicle condition data stored in the memory, command the material handling vehicle to move forward.
(5) According to some aspects of the present disclosure, a system for a remotely controllable material handling vehicle is provided. The material handling vehicle is operable in a manual mode where an operator can maneuver the material handling vehicle normally and a travel request mode where the operator can remotely request the material handling vehicle to move forward. The system includes a controller including a processor and a memory and a remote control device in communication with the controller. The controller is configured to continuously receive vehicle condition data to be stored in the memory, receive a travel request from the remote control device, recall stored vehicle condition data from the memory, and based on the stored vehicle condition data, command the material handling vehicle to move forward.
(6) According to some aspects of the present disclosure, a method for remotely operating a material handling vehicle in a travel request mode is provided. The method can include receiving a travel request from a remote control device in communication with the material handling vehicle, recalling at least one previously evaluated vehicle condition to determine if the at least one vehicle condition is appropriate for moving the material handling vehicle forward, and, upon the determination that the at least one vehicle condition is appropriate, commanding the material handling vehicle to move forward.
(7) The foregoing and other aspects and advantages of the disclosure will appear from the following description. In the description, reference is made to the accompanying drawings which

form a part hereof, and in which there is shown by way of illustration a preferred configuration of the disclosure. Such configuration does not necessarily represent the full scope of the disclosure, however, and reference is made therefore to the claims and herein for interpreting the scope of the disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The invention will be better understood and features, aspects and advantages other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such detailed description makes reference to the following drawings.

(2) FIG. **1** is a top, front, left isometric view of a non-limiting example of a material handling vehicle according to aspects of the present disclosure.

(3) FIG. **2** is a top, left side view of the material handling vehicle of FIG. **1**.

(4) FIG. **3** is a schematic illustration of a system for a remotely controllable material handling vehicle according to aspects of the present disclosure.

(5) FIG. **4** is a schematic illustration of a continuous evaluation loop according to aspects of the present disclosure.

(6) FIG. **5** is a schematic illustration of a method of switching a mode of operation of the material handling vehicle.

(7) FIG. **6** is a schematic illustration of a method of operation of the material handling vehicle in a travel request mode.

(8) FIG. **7** is an exemplary illustration of an operator-controlled remote control device in communication with the material handling vehicle of FIG. **1**.

DETAILED DESCRIPTION

(9) Before any aspect of the present disclosure are explained in detail, it is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The present disclosure is capable of other configurations and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

(10) It is also to be understood that any reference to an element herein using a designation such as “first,” “second,” and so forth does not limit the quantity or order of those elements, unless such limitation is explicitly stated. Rather, these designations may be used herein as a convenient method of distinguishing between two or more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements may be employed there or that the first element must precede the second element in some manner.

(11) The following discussion is presented to enable a person skilled in the art to make and use aspects of the present disclosure. Various modifications to the illustrated configurations will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other configurations and applications without departing from aspects of the present disclosure. Thus, aspects of the present disclosure are not intended to be limited to configurations shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The

following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected configurations and are not intended to limit the scope of the present disclosure. Skilled artisans will recognize the non-limiting examples provided herein have many useful alternatives and fall within the scope of the present disclosure.

(12) It is to be appreciated that material handling vehicles are designed in a variety of configurations to perform a variety of tasks. The various configurations of material handling vehicles described herein are shown by way of example. It will be apparent to those of skill in the art that the present invention is not limited to vehicles of these types, and can also be provided in various other types of material handling vehicle configurations, including for example, order pickers, reach vehicles, counterbalanced vehicles, and any other material handling vehicles. The various aspects disclosed herein are suitable for all of driver controlled, pedestrian controlled, remotely controlled, and autonomously controlled material handling vehicles.

(13) FIGS. **1** and **2** illustrate one non-limiting example of a material handling vehicle **100** ("MHV") according to the present disclosure. The material handling vehicle **100** may include a vehicle frame **102**, a traction wheel **104**, a power section **108**, and an operator compartment **110**. The power section **108** may be disposed within the vehicle frame **102** and may include a battery (not shown) configured to supply power to various components of the material handling vehicle **100**. For example, the battery may supply power to a motor (not shown) and/or transmission (not shown) disposed within the power section **108** and configured to drive the traction wheel **104**. In the illustrated non-limiting example, the traction wheel **104** is arranged under the power section **108**. In other non-limiting examples, the traction wheel **104** may be arranged in another location under the vehicle frame **102**.

(14) The operator compartment **110** may include a control handle **111** configured to provide a user interface for an operator and to allow the operator to control a speed and direction of travel of the material handling vehicle **100**. In some non-limiting examples, the control handle **111** may be configured to manually steer and control power to the traction wheel **104**.

(15) In the illustrated non-limiting example shown in FIGS. **1** and **2**, the material handling vehicle **100** includes the operator compartment **110** arranged rearward of the power section **108** and having an operator opening **112** that opens towards lateral sides **114** of the material handling vehicle **100**. The operator compartment **110** may also contain a floor mat **116** on which an operator of the material handling vehicle **100** may stand. In some non-limiting examples, the material handling vehicle **100** may be designed with the operator compartment arranged differently, for example, with an operator opening **112** that opens rearwardly. In the illustrated non-limiting example, the material handling vehicle **100** includes a pair of forks **118** that can be raised or lowered via actuators (not shown) in response to commands from the control handle **111**.

(16) In the illustrated non-limiting example shown in FIG. **2**, the material handling vehicle **100** may also include a holster **119**. The holster **119** can, for example, be attached to a portion of the material handling vehicle **100**. In the illustrated non-limiting example, the holster **119** is attached to a lateral side **114** of the material handling vehicle **100**. The holster **119** can be configured to temporarily hold or store a remote control device **122** when the remote control device **122** is not being held by the operator. In some embodiments, the holster **119** can include an integrated charger **121** for the remote control device **122** such that the remote control device **122** can be charged while being stored in the holster **119**. In some non-limiting examples, when an operator places the remote control device **122** in the holster **119**, the remote control device **122** can be unpaired from the material handling vehicle **100**.

(17) FIG. **3** illustrates one non-limiting example of a system **120** for a remotely controllable material handling vehicle, which may be implemented, for example, in the material handling vehicle **100** shown in FIG. **1**. The system **120** includes a remote control device **122** in communication with the material handling vehicle **100**. The remote control device **122** is operable

by an operator of the material handling vehicle **100** and can include a travel control function **124** and an I/O function **126**. The travel control function **124** and the I/O function **126** can be configured as first and second manually operable functions that can be operated by, for example, a button or a switch located on the remote control device **122**. For example, the travel control function **124** can be operated by a first button **125** on the remote control device **122** and the I/O function **126** can be operated by a second button **127** on the remote control device **122** (see FIG. **7**). The travel control function **124** and the I/O function **126** can be in electrical communication with a transmitter **128** within the remote control device **122**. The transmitter **128** on the remote control device **122** can wirelessly communicate with the material handling vehicle **100** via a receiver **130** located on the material handling vehicle **100**, as represented by dashed line **132**. In some non-limiting examples, the remote control device **122** may turn off after several minutes of inactivity.
(18) As will be described herein, the travel control function **124** can be configured to send a first signal from the transmitter **128** to the receiver **130** located on the material handling vehicle **100**. The first signal can be configured to instruct the material handling vehicle **100** to move forward from one location to the next within a warehouse while the operator is operating the travel control function **124** on the remote control device **122**. For example, the operator may desire the material handling vehicle **100** to move in an aisle along a rack within a warehouse from one picking location to the next picking location without the operator frequently entering/exiting the operator compartment **110** on the material handling vehicle **100**. Instead, the operator can operate the travel control function **124** on the remote control device **122** from outside the operator compartment **110** to move the material handling vehicle **100**. That is, the travel control function **124**, when actuated by the operator, provides a request to the material handling vehicle **100** to move forward. In some non-limiting examples, a duration of the forward movement of the material handling vehicle can be controllable by maintaining the travel control function **124** in an actuated state. For example, the operator can control the duration of time the material handling vehicle **100** is moving by depressing and holding the travel control function **124**. In some non-limiting examples, releasing the travel control function **124** after the travel control function **124** has been held can cause the material handling vehicle **100** to stop moving forward and/or coast to a stop.
(19) In some non-limiting examples, there may be a limited time duration on the holding of the travel request function **124** (e.g., after ten seconds, or more or less than ten seconds), after which travel of the material handling vehicle **100** may stop. In this specific non-limiting example, if the operator needs to “reset” this limited time duration, a rapid “release-and-re-hold” (e.g., within 2.5 seconds, or more or less than 2.5 seconds) of the travel request function **124** can “reset” the limited time duration without the material handling vehicle **100** coming to a stop.
(20) The I/O function **126** can be configured to send a second signal from the transmitter **128** to a receiver **130** located on the material handling vehicle **100**. In some non-limiting examples, the second signal may cause a series of events to occur, such as unpairing the remote control device **122** from the material handling vehicle **100** and stopping the material handling vehicle **100** from moving forward. For example, the I/O function **126**, when actuated by the operator, unpairs the remote control device **122** from the material handling vehicle **100**.
(21) The material handling vehicle **100** can also include a vehicle manager **134** (“VM”). In the illustrated non-limiting example, the vehicle manager can be in electrical communication with a steering control system **136** and a traction controller **138**. The vehicle manager **134** can be configured to issue commands and control the steering control system **136**. For example, the vehicle manager **134** can send a signal to the steering control system **136** to perform a steering maneuver or issue steering commands. The steering control system **136** may then perform the issued steering commands based on the received signal. The vehicle manager **134** can be configured to issue commands and control the traction controller **138**. For example, the vehicle manager **134** can send a signal to the traction controller **138** to move, position, accelerate, slow, or otherwise change a speed of the material handling vehicle **100**. The traction controller **138** may

then communicate (e.g., via electrical communication) to a brake **140** or a power unit **142** to change the speed of the material handling vehicle. In addition, the vehicle manager **134** can be configured to evaluate vehicle condition data from the steer control system **136** and the traction controller **138**. In one non-limiting example, the steer control system **136** can sense (e.g., via a sensor) a steering angle of the material handling vehicle **100**. In another non-limiting example, the traction controller **138** can sense a speed (e.g., via a sensor) of the material handling vehicle **100**. In either case, the sensed steering angle or speed of the material handling vehicle **100** can be communicated to the vehicle manager **134**.

(22) Referring still to FIG. **3**, the material handling vehicle **100** may also include a travel request controller **144** ("TRC"). The travel request controller **144** can have a memory **146** and a processor **148**. The memory **146** can be configured to store a plurality of executable instructions that may be carried out by the processor **148**. In addition, as will be described herein, the processor **148** can execute a continuous evaluation loop to continuously evaluate and store, for example, vehicle condition data of the material handling vehicle **100** in the memory **146**. The processor **148** can be configured to interpret and perform calculations using the vehicle condition data stored on the memory **146**.

(23) The travel request controller **144** can be in electrical communication with the receiver **130** such that the receiver **130** can receive the first and second signals from the transmitter **128** on the remote control device **122** and transmit the first and second signals to the travel request controller **144** to interpret them. As will be described herein, the travel request controller **144** can also be in electrical communication with the vehicle manager **134** such that signals, commands, and vehicle data can be sent or communicated between the travel request controller **144** and the vehicle manager **134**. For example, the travel request controller **144** can continuously (e.g., about every 20 milliseconds, or more or less) communicate vehicle command information to the vehicle manager **134**. In one non-limiting example, the vehicle command information can be transmitted from the travel request controller **144** to the vehicle manager **134** via a process data object ("PDO") message. The vehicle command information continuously being delivered from the travel request controller **144** can include travel commands (e.g., a request for forward vehicle motion), fork **118** raising or lowering commands, horn activation commands, stop commands, and control mode requests (e.g., a request to transition to/from a manual mode or a travel request mode). In addition, when the material handling vehicle **100** is in a travel request mode, the travel request controller **144** may send a steering command to the vehicle manager **134**, or the travel request controller **144** can set a maximum vehicle speed limitation or acceleration limitation through the vehicle manager **134** (e.g., limiting the maximum speed of the material handling vehicle to a walking speed during remote operation, etc.).

(24) The material handling vehicle **100** can also have a sensor **150** (e.g., positioned at a front of the vehicle, see FIGS. **1-2**). In one non-limiting example, the sensor **150** can be a time-of-flight camera or a 2D or 3D LIDAR scanner, however other obstacle detection capable sensors are also envisioned, such as any other obstacle detection sensor known in the art. The sensor **150** can be in electrical communication with the travel request controller **144** and can be configured to sense or detect obstacles located in or near a travel path of the vehicle. In one non-limiting example, the sensor **150** may also be configured to detect a rack located along a lateral side **114** of the material handling vehicle **100**.

(25) Referring now to FIGS. **1-3**, the material handling vehicle **100** can be configured to transition between a manual mode and a travel request mode. In one non-limiting example, the material handling vehicle **100** can have a switch **151** (e.g., a manually operable switch, see FIG. **2**) that can be positioned within or near the operator compartment **110** for access by the operator of the material handling vehicle **100**. For example, the switch can be positioned on the control handle **111**, next to a display **115**, or on a control panel **117** and accessible by the operator. In another non-limiting example, the switch can be positioned on the outside of the material handling vehicle **100**.

For example, the switch can be positioned on one or both of the lateral sides **114** or on the vehicle frame **102** of the material handling vehicle **100**. In another non-limiting example, the transition between a manual mode and a travel request mode can be initiated by the travel request controller **144** upon a pairing of the remote control device **122** to the material handling vehicle **100**, or upon activating the travel control function **124**.
(26) Now that the components of the system **120** have been described, operation of the remotely controlled material handling vehicle **100** will be described in the paragraphs to follow with reference to FIGS. **1-3**. When the material handling vehicle **100** is in manual mode, the material handling vehicle **100** can be operated normally by the operator. For example, the operator can use the control handle **111** on the material handling vehicle **100** to accelerate, decelerate, steer, or otherwise maneuver the material handling vehicle **100** manually by using a throttle or brake button/lever located on the control handle **111**. In one non-limiting example, the control handle **111** may include one or more jog buttons **113** configured to, when engaged, cause the material handling vehicle **100** to travel at a walking speed. Upon release of the jog button **113**, the material handling vehicle **100** may coast to a stop.
(27) The remote control device **122** can be paired, unpaired, or otherwise connected/disconnected to the material handling vehicle **100** when the material handling vehicle **100** is in the manual mode. However, travel or I/O commands (e.g., the first and second signals, respectively) generated by the remote control device **122** may not be executed by travel request controller **144** or communicated to the vehicle manager **134**. For example, when the material handling vehicle **100** is in the manual mode, the operator is unable to maneuver the material handling vehicle **100** using the remote control device **122**. According to some embodiments, I/O commands delivered by the remote control device **122** may still be sent to the travel request controller **144** to unpair the remote control device **122**.
(28) Conversely, when the material handling vehicle **100** is in the travel request mode, travel or I/O signals (e.g., the first and second signals, respectively) generated by the remote control device **122** are communicated to the vehicle manager **134** via the travel request controller **144**. For example, when the material handling vehicle **100** is in the travel request mode, the operator can cause the material handling vehicle **100** to move forward using the travel control function **124** and can cause the material handling vehicle **100** to unpair from the remote control device **122** using the I/O function **126** on the remote control device **122**. Additionally, when the material handling vehicle **100** is in the travel request mode, the operator is unable to maneuver the material handling vehicle **100** using the control handle **111**. For example, steering or throttle inputs, including inputs from the jog buttons **113**, from the control handle **111** may not be communicated to, or ignored by, the vehicle manager **134**. For example, the throttle on the control handle **111** can be prevented from controlling the speed of the material handling vehicle **100**. In some embodiments, the jog buttons **113** may remain functional. For example, when the material handling vehicle **100** is in the travel request mode position, and the remote control device **122** is yet to be paired, the jog buttons **113** may remain functional. If a remote control device **122** is then paired to the material handling vehicle **100**, the jog buttons **113** may no longer be functional. In some non-limiting examples, a horn button (not shown) may be located on the control handle **111**, and the horn button may remain active and usable by the operator when the material handling vehicle **100** is in either of the manual mode or the travel request mode. The operator may also be unable to manipulate the forks **118** using the control handle **111** when the material handling vehicle **100** is in the travel request mode. For example, when the remote control device **122** is paired with the material handling vehicle **100**, the forks **118** may not be maneuverable (e.g., raised or lowered) by the operator.
(29) Referring now to FIG. **4**, an exemplary schematic of the continuous evaluation loop **200** is illustrated. As previously described herein, the continuous evaluation loop **200** may be executed continuously, for example, by processor **148**, at predetermined discrete time intervals (e.g., about every 20 milliseconds to about every 1 second, or more than 1 second or less than 20 milliseconds)

while the material handling vehicle **100** is in operation (e.g., while the material handling vehicle **100** is “ON”, irrespective of the mode the vehicle is in). The continuous evaluation loop **200** can start when the travel request controller **144** receives vehicle condition data from the vehicle manager **134**, as illustrated by input block **202**. In one non-limiting example, the vehicle condition data can be communicated from the vehicle manager **134** to the travel request controller **144** via a process data object. In one non-limiting example, the vehicle condition data can include one or both of the steering angle and the vehicle speed of the material handling vehicle **100**. For example, the steer control system **136** can sense the steering angle of the material handling vehicle **100** and the traction controller **138** can sense the speed of the material handling vehicle **100**. The sensed steering angle and speed of the material handling vehicle **100** can be communicated to the vehicle manager **134**, and the vehicle manager **134** can then communicate the vehicle condition data to the travel request controller **144**.

(30) In another non-limiting example, the vehicle condition data communicated from the vehicle manager **134** to the travel request controller **144** can further include the current mode the material handling vehicle **100** is operating in (e.g., either the manual mode or the travel request mode). In another non-limiting example, the vehicle condition data can further include brake application status. For example, the vehicle manager **134** can communicate through the traction controller **138** to determine a status of the brake **140**, and communicate that status to the travel request controller **144**.

(31) At step **204**, the travel request controller **144** can store the vehicle condition data (e.g., steering angle, vehicle speed, etc.) on the memory **146** in the travel request controller **144**. At step **206**, the travel request controller **144** may then determine a vehicle condition, such as a motion condition, of the material handling vehicle. For example, the travel request controller **144** can determine if the material handling vehicle **100** is stopped. For example, the travel request controller **144** may use the vehicle speed received from the vehicle manager **134** or the vehicle speed stored in the memory **146** and compare the vehicle speed to a threshold value defined for a stopped vehicle (e.g., approximately 0 mph) to determine the motion condition of the material handling vehicle **100**. In one non-limiting example, the threshold value can be zero mph. If the travel request controller **144** determines that the vehicle speed is greater than the threshold value, then the travel request controller **144** stores (e.g., in the memory **146**) that the material handling vehicle **100** is “IN MOTION” (or some other representative designation that may indicate that the vehicle is not stopped), as indicated by step **208**. Conversely, if the travel request controller **144** determines that the vehicle speed is at or less than the threshold value, then the travel request controller **144** stores that the material handling vehicle **100** is “STOPPED” (or some other representative designation that may indicate that the vehicle is stopped), as indicated by step **210**.

(32) As previously noted, the continuous evaluation loop **200** may run continuously at discrete time intervals, which can enable, for example, the memory **146** of the travel request controller **144** to have recent and accurate vehicle condition data stored on the memory **146**. In one non-limiting example, the continuous evaluation loop **200** can be executed by the travel request controller **144** about every 20 milliseconds. In another non-limiting example, the continuous evaluation loop **200** can be executed about every second, any time period between 20 milliseconds and one second, or any time period more than 1 second or less than 20 milliseconds. One of ordinary skill in the art readily recognizes that the specific time period chosen can be application specific and is not intended to be limiting in any way.

(33) As will be described in greater detail in the paragraphs to follow, the travel request controller **144** can execute the continuous evaluation loop **200** such that the travel request controller **144** can evaluate the most recent vehicle condition data stored in the memory **146** upon receiving a travel command or a stop command given by the travel control function **124** or the I/O function **126** on the remote control device **122**. In one non-limiting example, only the most recent vehicle condition data can be stored in the memory **146**. For example, one data point can be stored and then

continuously updated (e.g., overwritten) in the memory **146** as the continuous evaluation loop **200** is executed by the travel request controller **144**. In another non-limiting example, a history of the vehicle condition data can be stored in the memory **146**.

(34) Referring now to FIG. **5**, a method **300** of switching the material handling vehicle **100** from the manual mode to the travel request mode will be described. The process may start at step **302**, where a request to transition from the manual mode to the travel request mode may be initiated (e.g., via one of the methods previously described herein). At step **304**, the travel request controller **144** may then recall at least one of the most recent vehicle condition data stored on the memory **146** (e.g., the vehicle condition data that was updated on the previous continuous evaluation loop **200** execution) and determine if the stored vehicle conditions are appropriate for transitioning from the manual mode to the travel request mode. In that way, the travel request controller **144** can use vehicle condition data and motion condition information that has been previously updated to determine if vehicle conditions are appropriate for transitioning modes. For example, the travel request controller **144** can ensure that the most recent motion condition stored in the memory **146** is “STOPPED.” In addition, in some non-limiting examples, the travel request controller **144** can also check other aspects of the material handling vehicle **100** at step **304**. For example, the travel request controller **144** can query the vehicle manager **134** to ensure that the material handling vehicle **100** is being operated in a manual mode. The travel request controller **144** can also check to ensure the I/O function **126** on the remote control device **122** is not currently being actuated by an operator. If the travel request controller **144** determines that the vehicle conditions are not appropriate for transitioning from the manual mode to the travel request mode, the travel request controller takes no action and continues to evaluate the vehicle conditions (e.g., by executing the continuous evaluation loop **200**).

(35) Upon the travel request controller **144** determining that the vehicle conditions are appropriate based on the previously stored vehicle condition data, the travel request controller **144** may then send a signal to the vehicle manager **134** to request a transition from the manual mode to the travel request mode at step **306**. In response to the signal sent from the travel request controller **144** to the vehicle manager **134**, the vehicle manager **134** can then enable the material handling vehicle to switch to the travel request mode at step **308**. With the material handling vehicle **100** in the travel request mode, remote operation of the material handling vehicle **100** by the operator using the remote control device **122** is then enabled. For example, the operator can use the remote control device **122** to send signals to the travel request controller **144**, as previously described herein. The travel request controller **144** can then generate vehicle commands and then communicate those vehicle commands to the vehicle manager **134** for execution.

(36) The material handling vehicle **100** can be switched back to the manual mode from the travel request mode by the operator (e.g., by actuating the switch **151**, see FIG. **1**). According to one non-limiting example, the material handling vehicle can be switched back to the manual mode from the travel request mode by the vehicle manager **134**, if the vehicle manager **134** detects vehicle conditions that result in switching back to the manual mode.

(37) Referring now to FIGS. **3** and **6**, a method **400** of remotely operating the material handling vehicle **100** using the remote control device **122** when the material handling vehicle **100** is in a travel request mode is illustrated. In the exemplary discussion to follow, the example of an order picking operation will be described, where an operator of the material handling vehicle **100** may desire to move in an aisle and along a rack within a warehouse from one picking location to the next picking location. One of ordinary skill in the art readily recognizes that this is just one example of operation and is not intended to be limiting in any way.

(38) The method **400** may begin at step **402**, where the operator can actuate one of the travel control function **124** or the I/O function **126** on the remote control device **122**, thus generating a first signal or a second signal, respectively. At step **404**, the transmitter **128** may wirelessly communicate the first and/or second signal to the travel request controller **144** via the receiver **130**

in communication with the transmitter **128**.

(39) If the operator actuates the travel control function **124**, the travel request controller **144** may then recall at least one of the most recent vehicle condition data stored on the memory **146** (e.g., the vehicle condition data that was updated on the previous/most recent continuous evaluation loop **200** execution) at step **406** and determine if the stored vehicle conditions are appropriate for executing the travel request issued by the remote control device **122**. For example, the travel request controller **144** can ensure that the most recent motion condition stored in the memory **146** is "STOPPED." In that way, the travel request controller **144** uses vehicle condition data and motion condition information that has been previously updated when determining if vehicle conditions are appropriate for the travel/stop command.

(40) At step **408**, upon determining that the vehicle conditions are appropriate based on the previously stored vehicle condition data, the travel request controller **144** may then send a request (e.g., a vehicle command) to the vehicle manager **134** based on the received signal at step **404**. For example, if the operator actuates the travel control function **124** on the remote control device **122** (e.g., by depressing the first button **125** on the remote control device **122**, see FIG. **7**), the travel request controller **144** can send a travel command to the vehicle manager **134** to instruct the material handling vehicle **100** to move, or otherwise travel forward. According to another example, if the operator actuates the I/O function **126** on the remote control device **122**, an unpair command is sent to the travel request controller **144** to request that the material handling vehicle **100** be unpaired from the remote control device **122** and a stop command can be sent from the travel request controller **144** to the vehicle manager **134** to instruct the material handling vehicle **100** to come to a stop.

(41) In response to the received commands, the vehicle manager **134** can communicate to the traction controller **138** at step **410** to operate the power unit **142** to enable the material handling vehicle **100** to travel or stop (e.g., if a travel command or a stop command is received, respectively). For example, the material handling vehicle **100** may be at a first picking location within an aisle of a warehouse and the operator may desire the material handling vehicle **100** to travel to a second picking location within that aisle. The travel command executed by the vehicle manager **134** can enable the material handling vehicle to travel forward from the first picking location to the second picking location.

(42) While the material handling vehicle **100** is traveling, the travel request controller may monitor the sensor **150** to determine if the material handling vehicle **100** should steer or stop. In the case that the sensor **150** provides an indication that the material handling vehicle **100** should steer, the travel request controller **144** can send a steer command to the vehicle manager **134** to request a steering maneuver. In response to the received steering command, the vehicle manager **134** can communicate to the steer control system **136** to maneuver the material handling vehicle **100** (e.g., to steer the material handling vehicle **100** to travel alongside a rack structure). In the case that the sensor **150** provides an indication that the material handling vehicle **100** should stop, for example, in response to detecting an obstacle in a travel path of the material handling vehicle **100**, the travel request controller can send a stop command to the vehicle manager **134** to request that the material handling vehicle **100** come to a stop. In response to the received stop command, the vehicle manager **134** can communicate to the traction controller **138** to stop the material handling vehicle **100** (e.g., to stop the material handling vehicle **100** if the sensor **150** detects an obstacle in a travel path of the material handling vehicle **100**).

(43) For certain types of vehicles there are training requirements imposed by various government agencies, laws, rules and regulations. For example, the United States Department of Labor Occupational Safety and Health Administration (OSHA) imposes a duty on employers to train and supervise operators of various types of material handling vehicles. Recertification every three years is also required. In certain instances, refresher training in relevant topics shall be provided to the operator when required. In all instances, the operator remains in control of the material handling

vehicle during performance of any actions. Further, a warehouse manager remains in control of the fleet of material handling vehicles within the warehouse environment. The training of operators and supervision to be provided by warehouse managers requires among other things proper operational practices including among other things that an operator remain in control of the material handling vehicle, pay attention to the operating environment, and always look in the direction of travel.
(44) Within this specification embodiments have been described in a way which enables a clear and concise specification to be written, but it is intended and will be appreciated that embodiments may be variously combined or separated without parting from the invention. For example, it will be appreciated that all preferred features described herein are applicable to all aspects of the invention described herein.
(45) Thus, while the invention has been described in connection with particular embodiments and examples, the invention is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein.
(46) Various features and advantages of the invention are set forth in the following claims.

Claims

1. A material handling vehicle, comprising: a controller comprising memory and processing circuitry, the processing circuitry configured to: receive vehicle condition data from components of the material handling vehicle, the vehicle condition data indicative of a speed of the material handling vehicle and a steering angle of the material handling vehicle; store the vehicle condition data in the memory; execute a continuous evaluation loop to continuously update the vehicle condition data stored in the memory at predetermined discrete time intervals that are between 20 milliseconds and 1 second; receive a travel request signal from a remote control device based on an input provided by an operator to the remote control device, the travel request signal commanding the material handling vehicle to move from a first location to a second location; recall the vehicle condition data stored in the memory to determine a current vehicle condition for the material handling vehicle; evaluate the travel request signal relative to the current vehicle condition to determine that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location; and responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, command at least one of the components of the material handling vehicle to operate the material handling vehicle in accordance with the travel request signal.
2. The material handling vehicle of claim 1, wherein: in the first mode, the material handling vehicle does not operate in accordance with travel request signals received from the remote control device; and in the second mode, the material handling vehicle does operate in accordance with travel request signals received from the remote control device.
3. The material handling vehicle of claim 1, wherein the processing circuitry is configured to determine the current vehicle condition by comparing the speed of the material handling vehicle to a threshold value.
4. The material handling vehicle of claim 3, wherein the current vehicle condition indicates that the material handling vehicle is stopped or that the material handling vehicle is in motion.
5. The material handling vehicle of claim 1, wherein, responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, the processing circuitry is configured to transition the material handling vehicle from a first mode to a second mode.
6. The material handling vehicle of claim 1, further comprising an obstacle detection sensor,

wherein the processing circuitry is configured to receive the vehicle condition data from the obstacle detection sensor, and the vehicle condition data is further indicative of a presence of an obstacle in or near a travel path of the material handling vehicle.

7. A system for remotely operating a material handling vehicle, comprising: a material handling vehicle controller on the material handling vehicle, the material handling vehicle controller comprising processing circuitry and memory; and a remote control device in communication with the material handling vehicle controller, the remote control device comprising a transmitter that sends a travel request signal to the material handling vehicle based on an input provided by an operator to the remote control device; wherein the processing circuitry of the material handling vehicle controller is configured to: receive vehicle condition data from components of the material handling vehicle, the vehicle condition data indicative of a speed of the material handling vehicle and a steering angle of the material handling vehicle; store the vehicle condition data in the memory; execute a continuous evaluation loop to continuously update the vehicle condition data stored in the memory at predetermined discrete time intervals that are between 20 milliseconds and 1 second; receive the travel request signal from the remote control device based on the input provided by the operator to the remote control device, wherein the travel request signal commands the material handling vehicle to move from a first location to a second location; recall the vehicle condition data stored in the memory to determine a current vehicle condition for the material handling vehicle; evaluate the travel request signal relative to the current vehicle condition to determine that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location; and responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, command at least one of the components of the material handling vehicle to operate the material handling vehicle in accordance with the travel request signal.

8. The system of claim 7, wherein the processing circuitry of the material handling vehicle controller is configured to determine the current vehicle condition by comparing the speed of the material handling vehicle to a threshold value.

9. The system of claim 8, wherein the current vehicle condition indicates that the material handling vehicle is stopped or that the material handling vehicle is in motion.

10. The system of claim 7, wherein the vehicle condition data is further indicative of a presence of an obstacle in or near a travel path of the material handling vehicle.

11. The system of claim 7, wherein the transmitter wirelessly sends the travel request signal to the material handling vehicle controller.

12. The system of claim 7, wherein responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, the processing circuitry of the material handling vehicle controller is configured to transition the material handling vehicle from a first mode to a second mode.

13. The system of claim 12, wherein: in the first mode, the material handling vehicle does not operate in accordance with travel request signals received from the remote control device; and in the second mode, the material handling vehicle does operate in accordance with travel request signals received from the remote control device.

14. The system of claim 7, wherein the remote control device comprises: a first physical button selectable by the operator to send the travel request signal to the material handling vehicle controller; and a second physical button that is selectable by the operator to pair the remote control device from the material handling vehicle controller.

15. A method for remotely operating a material handling vehicle, comprising: receiving vehicle condition data from components of the material handling vehicle, the vehicle condition data indicative of a speed of the material handling vehicle and a steering angle of the material handling vehicle; storing the vehicle condition data in memory; executing a continuous evaluation loop to continuously update the vehicle condition data stored in the memory at predetermined discrete time

intervals, wherein the predetermined discrete time intervals are between 20 milliseconds and 1 second; receiving a travel request signal from a remote control device based on an input provided by an operator to the remote control device, the travel request signal commanding the material handling vehicle to move from a first location to a second location; recalling the vehicle condition data stored in the memory to determine a current vehicle condition for the material handling vehicle; evaluating the travel request signal relative to the current vehicle condition to determine that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location; and responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, commanding at least one of the components of the material handling vehicle to operate the material handling vehicle in accordance with the travel request signal.

16. The method of claim 15, wherein determining the current vehicle condition comprises comparing the speed of the material handling vehicle to a threshold value.

17. The method of claim 16, wherein the current vehicle condition indicates that the material handling vehicle is stopped or that the material handling vehicle is in motion.

18. The method of claim 15, further comprising, responsive to determining that the current vehicle condition is for commanding the material handling vehicle to move from the first location to the second location, transitioning the material handling vehicle from a first mode to a second mode.

19. The method of claim 18, wherein: in the first mode, the material handling vehicle does not operate in accordance with travel request signals received from the remote control device; and in the second mode, the material handling vehicle does operate in accordance with travel request signals received from the remote control device.

20. The method of claim 15, wherein the vehicle condition data is further indicative of a presence of an obstacle in or near a travel path of the material handling vehicle.
